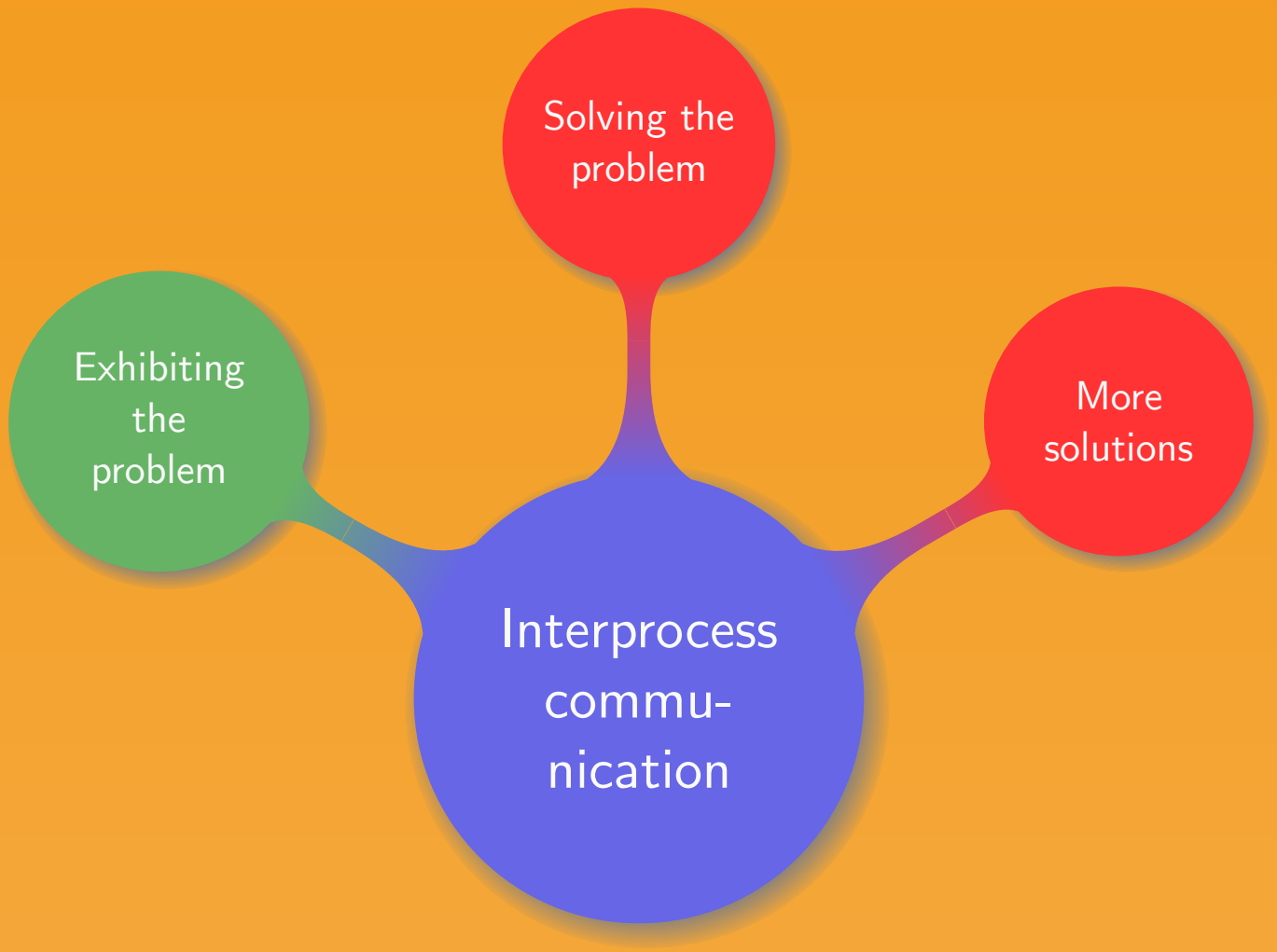
Introduction to Operating Systems

3. Interprocess communication

Manuel – Fall 2019

Chapter organisation



Easy setup

Independent threads:

- Cannot affect/be affected by anything
- State not shared with other threads
- Input state determines the output
- Reproducible
- No side effect when stopping/resuming

Executing threads

Where difficulties start:

- Single-tasking: run a thread to completion and start next one
- Multi-tasking:
  - One core shared among several threads
  - Several cores run several threads in parallel
- A thread runs on one core at a time
- A thread can run on different cores at different times
- For threads no difference between one or more cores

Cooperating threads

Setup:

- Threads sharing a state
- Behavior depends on the execution sequence
- Behavior may seem random/irreproducible

thread运行的顺序对变量产生影响, eg
thread1: a=3
thread2: a=4

Cooperating threads

Setup:

- Threads sharing a state
- Behavior depends on the execution sequence
- Behavior may seem random/irreproducible

Major problems:

1. How can threads/processes share information?
2. How to prevent them for getting in each other's way?
3. How to handle sequencing?

Atomic operation

Operation that either happens in its entirety or not at all:

- Atomic operation cannot be created
- Atomic operations are hardware level
- Central to the question of parallel programming

A=B: load B, save to A (2 operations)
atomic: the 2 operations are undivisible

Atomic operation

Operation that either happens in its entirety or not at all:

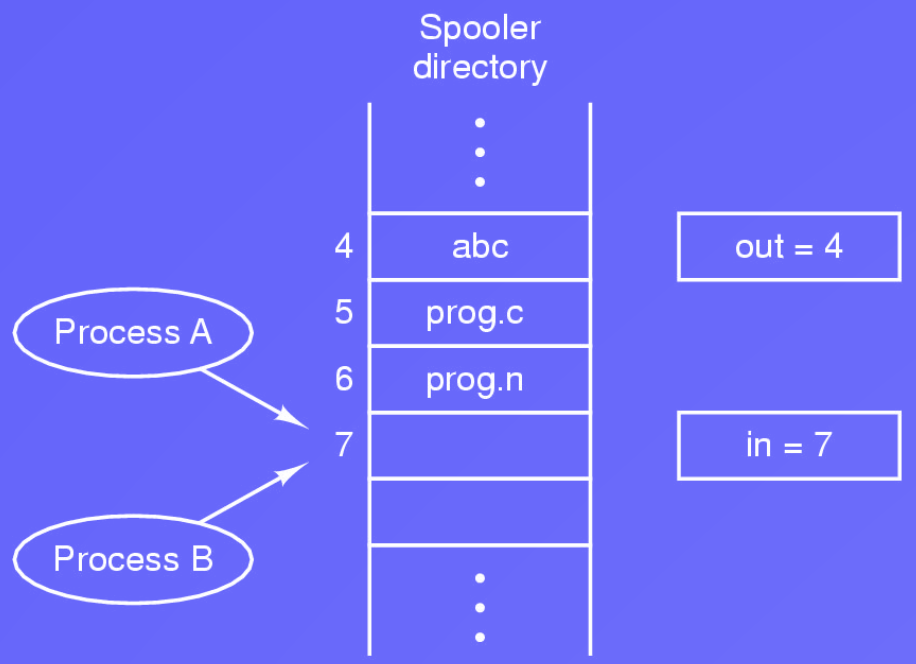
- Atomic operation cannot be created
- Atomic operations are hardware level
- Central to the question of parallel programming

Example.
Most basic atomic operation, `A=B`:

- Read a clean value for B
- Set a clean value for A

Race conditions

2 processes competing to read/write
-> global virable issue



Practical example:

- Process A wants to queue a file, reads next_free_slot=7
- Interrupt occurs, Process B reads next_free_slot=7
- Process B queues its file in slot 7, and update next_free_slot=8
- Process A resumes using next_free_slot=7
- Game over

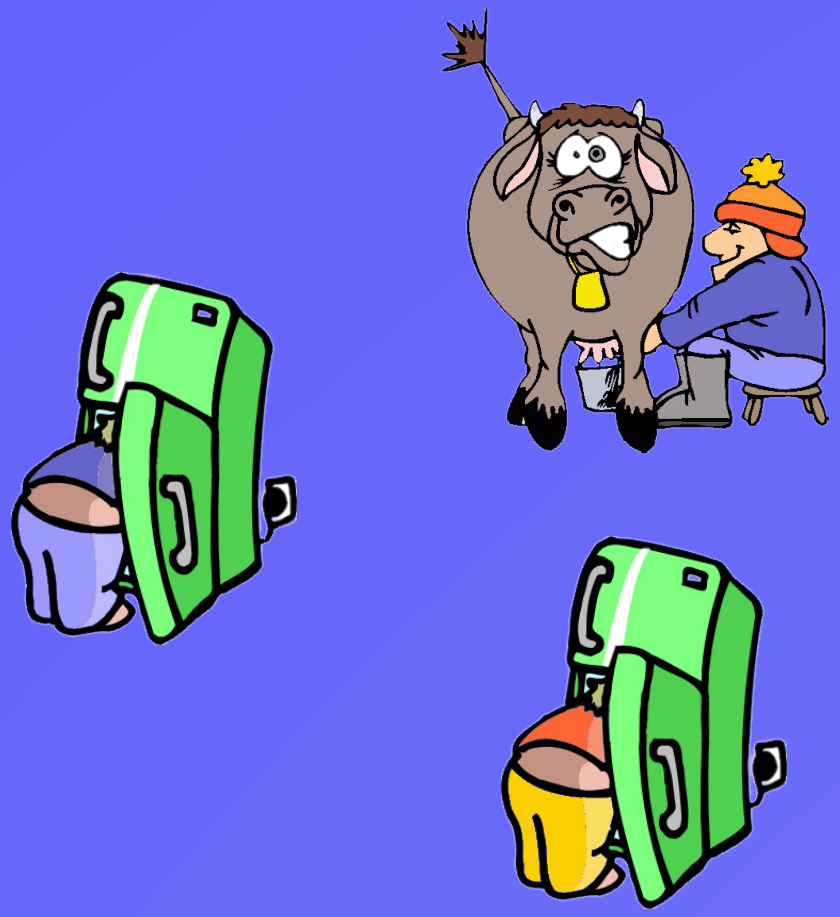
Too much milk

9:00 am	9:15 am	9:30 am	9:45 am

Too much milk

9:00 am	9:15 am	9:30 am	9:45 am

Too much milk



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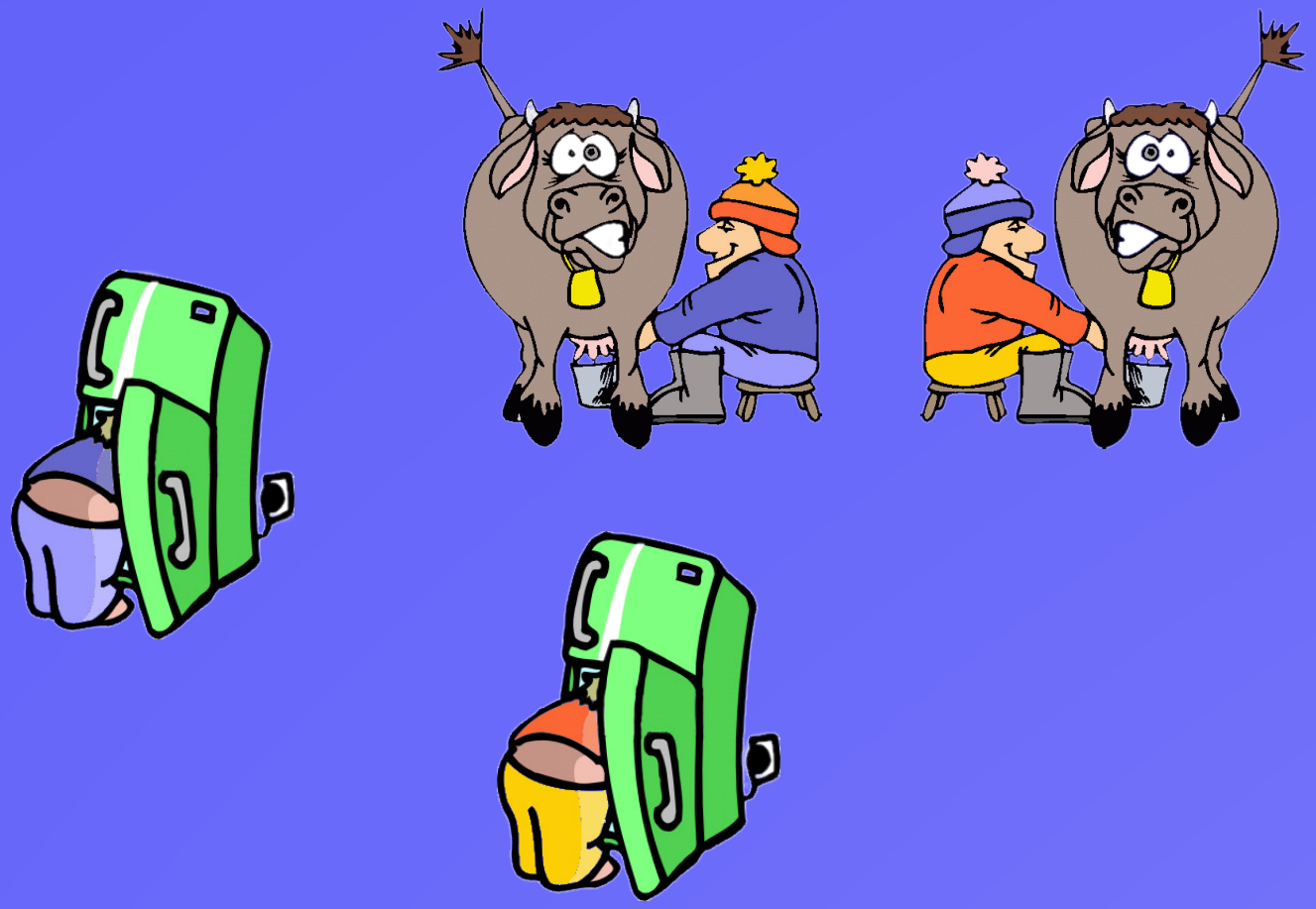
Too much milk



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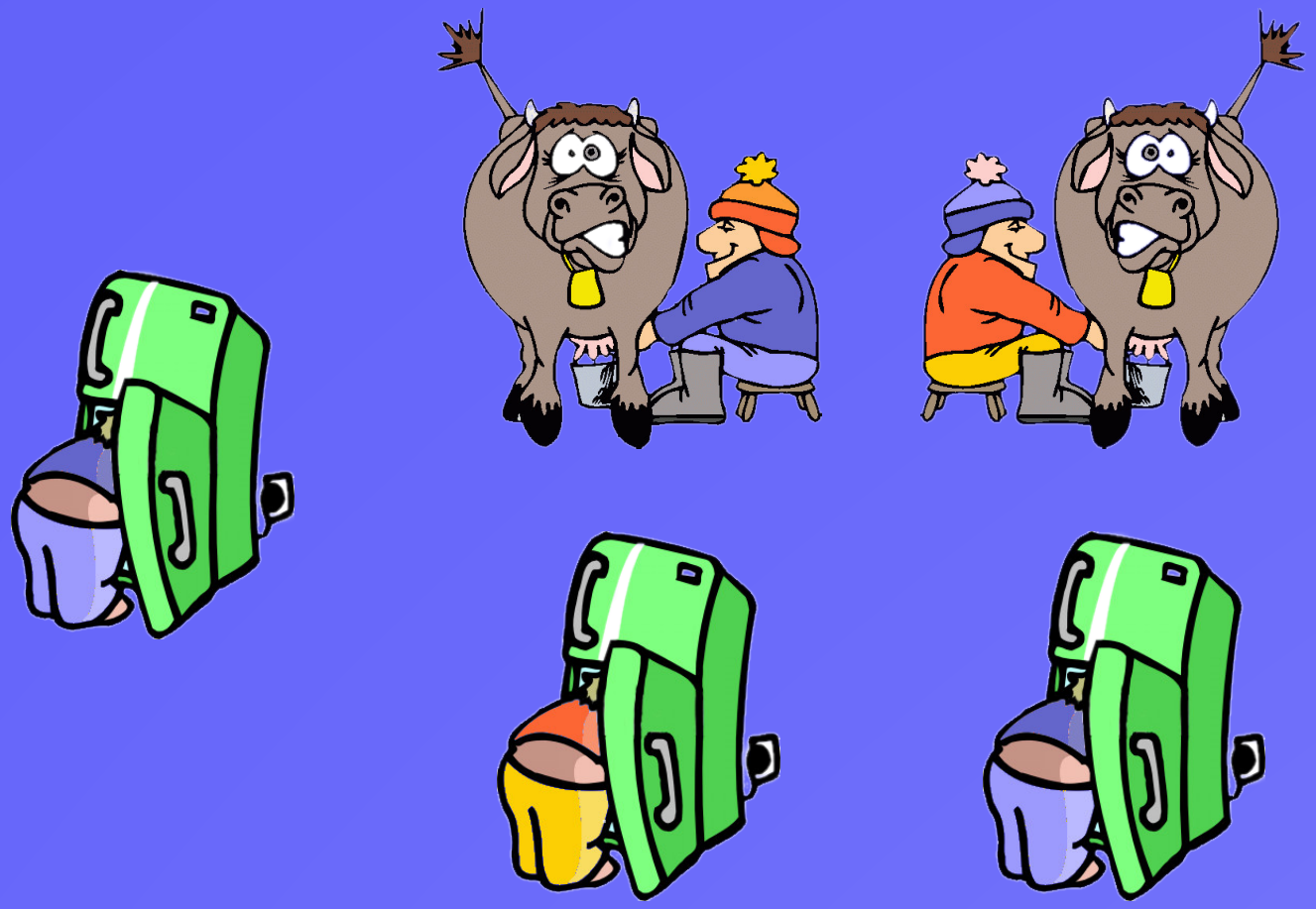

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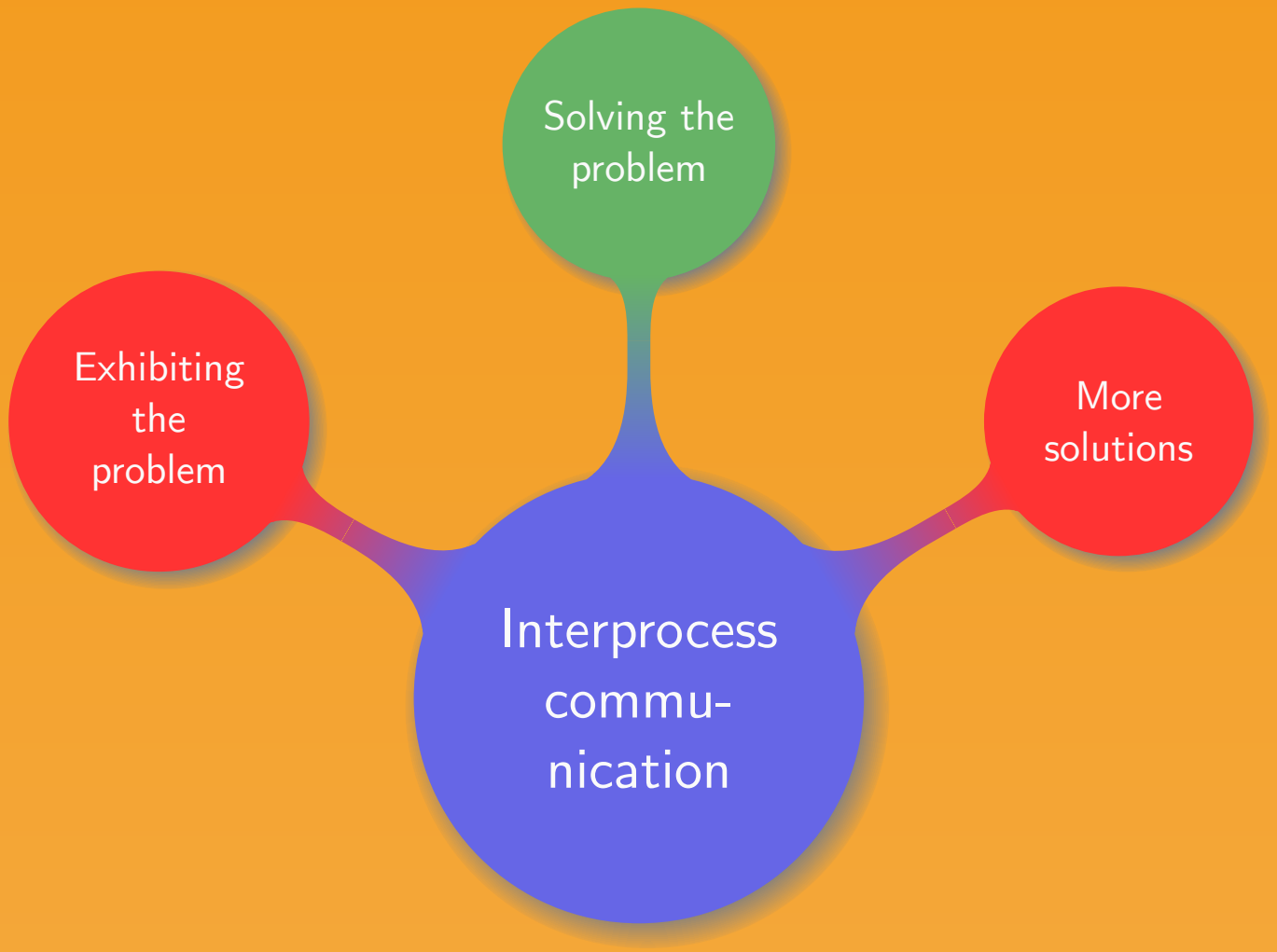
Chapter organisation



Critical region

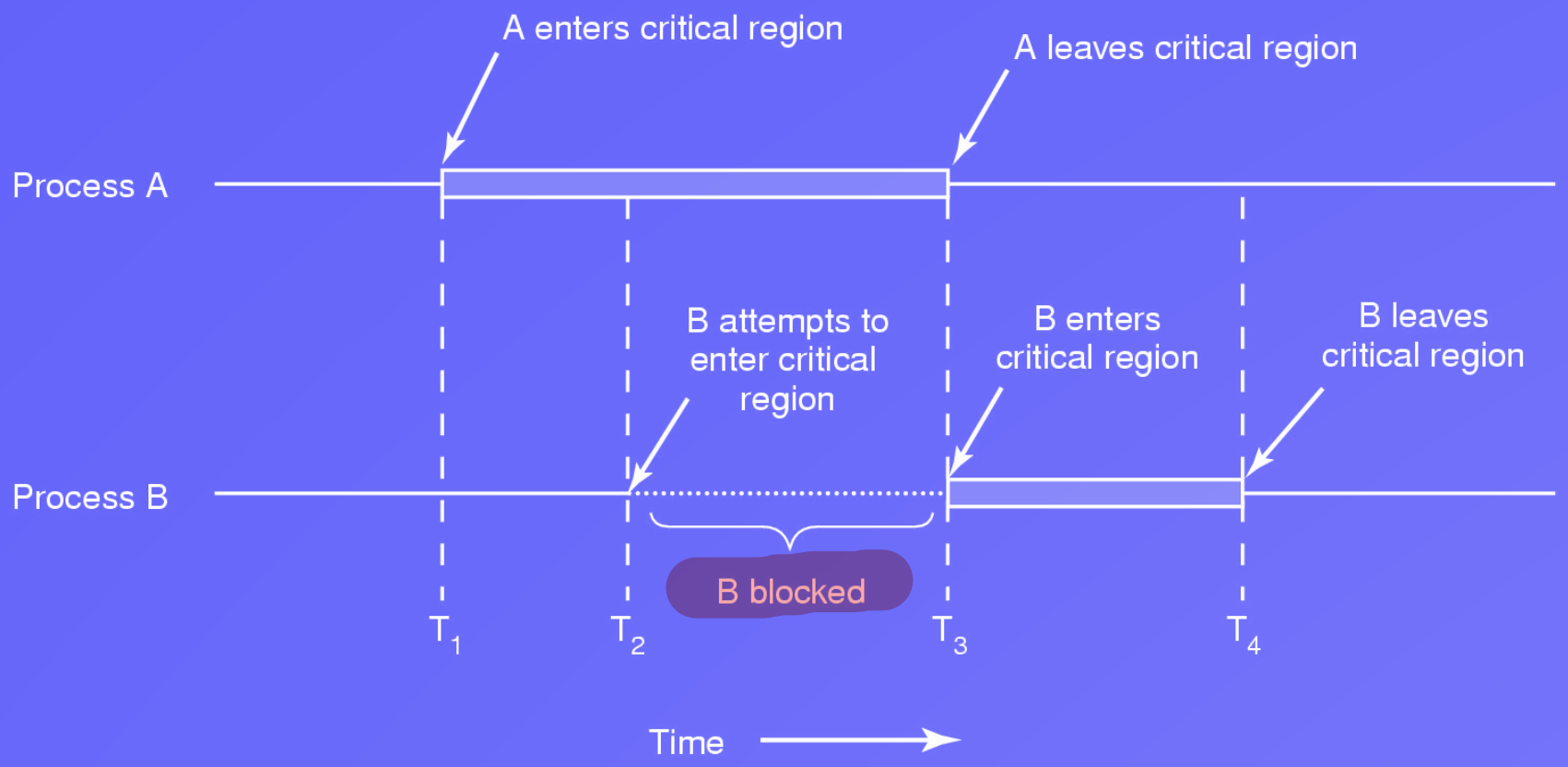
The fridge is the critical region. Previous process might chage the status of the fridge, so the next should be blocked from fridge

Part of the program where shared memory is accessed:

- No two processes can be in a critical region at the same time
- No assumption on speed/number of CPU
- No process outside a critical region can block other processes
- No process waits forever to enter its critical region

Mutual exclusion

need access to keyboard/ other input (eg hard drive)/ need access to critical region but occupied: blocked



Get interrupted: ready.

Too much milk

Frank

```
if(no milk && no note) {
  leave note;
  milk the cow;
  remove note;
}
```

John

```
if(no milk && no note) {
  leave note;
  milk the cow;
  remove note;
}
```

1 cpu, 2 threads

Too much milk

Frank

```
if(no milk && no note) {
  leave note;
  milk the cow;
  remove note;
}
```

John

```
if(no milk && no note) {
  leave note;
  milk the cow;
  remove note;
}
```

Result: too much milk

Too much milk

Frank

```
leave note Frank;
if(no note John) {
  if(no milk){
    milk the cow;
  }
}
remove note Frank;
```

John

```
leave note John;
if(no note Frank) {
  if(no milk) {
    milk the cow;
  }
}
remove note John;
```

frank leaves note -> swtich -> John leaves note & if falses -> switch -> frank if fails & remove note , end -> John remove note

Too much milk

Frank

```
leave note Frank;
if(no note John) {
  if(no milk){
    milk the cow;
  }
}
remove note Frank;
```

John

```
leave note John;
if(no note Frank) {
  if(no milk) {
    milk the cow;
  }
}
remove note John;
```

Result: no milk

Too much milk

Frank

```
leave note Frank;
while(note John) {
  nothing;
}
if(no milk) {
  milk the cow;
}
remove note Frank;
```

John

```
leave note John;
if(no note Frank) {
  if(no milk) {
    milk the cow;
  }
}
remove note John;
```

drawback:
not symmetry -> copmlicates code
Frank occupying cpu doing nothing

Too much milk

Frank

```
leave note Frank;
while(note John) {
  nothing;
}
if(no milk) {
  milk the cow;
}
remove note Frank;
```

John

```
leave note John;
if(no note Frank) {
  if(no milk) {
    milk the cow;
  }
}
remove note John;
```

Result: just enough milk

Too much milk

Comments on the solution:

- Solution is correct
- Complicated
- Asymmetrical
- Busy wait, CPU time wasted

Peterson's idea

Simple strategy:

- Assume two processes
- Two functions:
  - `enter_region`: show process is interested and wait for its turn
  - `leave_region`: indicates departure from critical region
- Solution is correct
- Drawback: busy wait

Peterson's idea

Simple strategy:

- Assume two processes
- Two functions:
  - `enter_region`: show process is interested and wait for its turn
  - `leave_region`: indicates departure from critical region
- Solution is correct
- Drawback: busy wait

Exercise.
Write the pseudo C code for Peterson's idea using two processes represented by the integers 0 and 1

Peterson's idea

```
1  #define TRUE 1
2  #define FALSE 0
3  int turn;
4  int interested[2];
5  void enter_region(int p) {
6    int other;
7    other=1-p;
8    interested[p]=TRUE;
9    turn=p;
10   while(turn==p && interested[other]==TRUE)
11 }
12 void leave_region(int p) {
13   interested[p]=FALSE;
14 }
```

turn指向下一个run的process

frank&john一起进line10, turn只指向其中一个

wait doing nothing

Peterson's idea

Side effects of Peterson's idea:

- Two processes: L, low priority, and H, high priority
- L enters in a critical region
- H becomes ready
- H has higher priority so the scheduler switches to H
- L has lower priority so is not rescheduled as long as H is busy
- H loops forever

Software vs. hardware?

Disabling interrupts, good or bad:

```
boolean flag[2];
int turn;
void procedure0()
{
while(true)
{
flag[0]=true;
turn=1;
while(flag[1]&&turn==1) /*若flag[1]为false，P0就进入临界区；若flag[1]为tureP0循环等待，只要P1退出临界区，P0即可进入*
{
/* donothing*/
}
visit();/*访问临界区*/
flag[0]=false;/*访问临界区完成，procedure0释放出临界区*/
/*remainder section*/
}
}
void procedure1()
{
while(true)
{
flag[1]=true;
turn=0;
while(flag[0]&&turn==0)
{
/* donothing*/ ;
}
visit();/*访问临界区*/
flag[1]=false;/*访问临界区完成，procedure1释放出临界区*/
/*remainder section*/
}
}
void main()
{
flag[0]=flag[1]=false;
/*start procedure0 and procedure1*/ ;
}
```

Software vs. hardware?

Disabling interrupts, good or bad:

- Case 1: block interrupts for the whole computer
  - Interrupts could be disabled for a while (too long?)
  - This gives a lot of power to the programmer
- Case 2: block interrupts for only one CPU
  - No effect on other processors
  - Another CPU can access the variable between read and write

The TSL instruction

A simple atomic operation:

- Test and Set Lock (TSL)
- Hardware level, requires assembly
- Task: copy *lock* to register and set *lock* to 1
- Atomic operation

The TSL instruction

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- Atomic operation

```
enter_region:
  TSL REGISTER,LOCK
  CMP REGISTER,#0
  JNE enter_region
  RET

leave_region:
  MOVE LOCK,#0
  RET
```

The TSL instruction

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  RET

leave_region:
  MOVE LOCK,#0
  RET
```

Note: TSL displays the same side-effect as Peterson's solution when dealing with processes having different priorities

Blocking strategy

```
#define N 100
int count=0;
void producer() {
  int item;
  while(1) {
    item=produce_item();
    if(count==N) sleep();
    insert_item(item); count++;
    if(count==1) wakeup(consumer);
  }
}
void consumer() {
  int item;
  while(1) {
    if(count==0) sleep();
    item=remove_item(); count--;
    if(count==N-1) wakeup(producer);
    consume_item(item);
  }
}
```

wait do nothing -> do other things

运行到这里， count=N，switch to consumer.
cunsumer remove everything
count = 0, sleep,
回去producer, producer sleep
两个线程同时sleep

Race condition

Assume the buffer is empty:

- Consumer reads `count == 0`
- Scheduler stops the consumer and starts the producer
- Producer adds one item
- Producer wakes up the consumer
- Consumer not yet asleep, signal is lost
- Consumer goes asleep
- When the buffer is full the producer falls asleep
- Both consumer and producer sleep forever

Semaphore

Basics:

- 1965, Edseger Dijkstra
- Simple hardware based solution
- Basis of all contemporary OS synchronization mechanisms

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A semaphore s in a non-negative variable that can only be changed or tested by two atomic actions:

```
down(s) {
  while(s==0) wait();
  s--;
}
```

```
up(s) {
  s++;
}
```

3个instruction
very short

Semaphore and processes

The down operation

Check if the value is > 0

- True: decrement the value and continues
- False: sleep, do not complete the down

Only one single atomic action

The up operation

- Increment the value of the semaphore
- If one or more processes were asleep, randomly choose one to wakeup (complete its down)

Only one single atomic action

Semaphore implementation

Semaphores MUST be implemented in an indivisible way:

- Up and down are implemented using system calls
- OS disables all interrupts while testing, updating the semaphore and putting process to sleep
- When dealing with more than one CPU, semaphores are protected using the TSL instruction

Note: a semaphore operation only takes a few microseconds

Semaphore and interrupts

Hiding interrupts using semaphores:

- Each I/O device gets a semaphore initialised to 0
- Managing process applies a `down` when starting an I/O device
- Process is blocked
- Interrupt handler applies an `up` when receiving an interrupt
- Process is ready to run again

Mutex

MUTual EXclusion:

- Simplified semaphore
- Takes values 0 (unlocked) or 1 (locked)

On a mutex-lock request:

- If mutex is currently unlocked, lock it; thread can enter in critical region
- If mutex is currently locked, block the calling thread until thread in critical regions is done and calls mutex-unlock
- Randomly chose which thread acquires the lock if more than one are waiting

Mutex implementation

Mutexes can be implemented in user-space using TSL

```
mutex-lock:
  TSL REGISTER,MUTEX
  CMP REGISTER,#0
  JNE ok
  CALL thread_yield
  JMP mutex-lock
ok: RET

mutex-unlock:
  MOVE MUTEX,#0
  RET
```

Questions.

- What differences were introduced compared to `enter_region` (3.20)?
- In user-space what happens if a thread tries to acquire lock through busy-waiting?
- Why is `thread_yield` used?

Example

consumer_producer.c

如果用if,wake up后可能有很多thread同时wake(broadcast signal)
但是只有一个thread可以给lock。用while可以verify again，
如果lock被别人拿走了就继续sleep

```
#include <stdio.h>
#include <pthread.h>
#define MAX 1000
pthread_mutex_t m; pthread_cond_t cc, cp; int buf=0;
void *prod() {
  for(int i=1;i<MAX;i++) {
    pthread_mutex_lock(&m); while(buf!=0) pthread_cond_wait(&cp,&m);
    buf=1; pthread_cond_signal(&cc); pthread_mutex_unlock(&m);
  }
  pthread_exit(0);
}
void *cons() {
  for(int i=1;i<MAX;i++) {
    pthread_mutex_lock(&m);  while(buf==0) pthread_cond_wait(&cc,&m);
    buf=0; pthread_cond_signal(&cp); pthread_mutex_unlock(&m);
  }
  pthread_exit(0);
}
int main() {
  pthread_t p, c;
  pthread_mutex_init(&m,0);  pthread_cond_init(&cc,0); pthread_cond_init(&cp,0);
  pthread_create(&c,0,cons,0); pthread_create(&p,0,prod,0);
  pthread_join(p,0); pthread_join(c,0);
  pthread_cond_destroy(&cc); pthread_cond_destroy(&cp); pthread_mutex_destroy(&m);
}
```

the one who locked the mutex can access the buf. Does not want 2 threads to access a global variable simultaneously

wait -> unlock the mutex
wait for buf to change to 0
another thread will wake it up

wake the consumer

wakeup后查看buf是不是0，
不是的话继续sleep，
是的话就可以把buf变成1，然后wake consumer

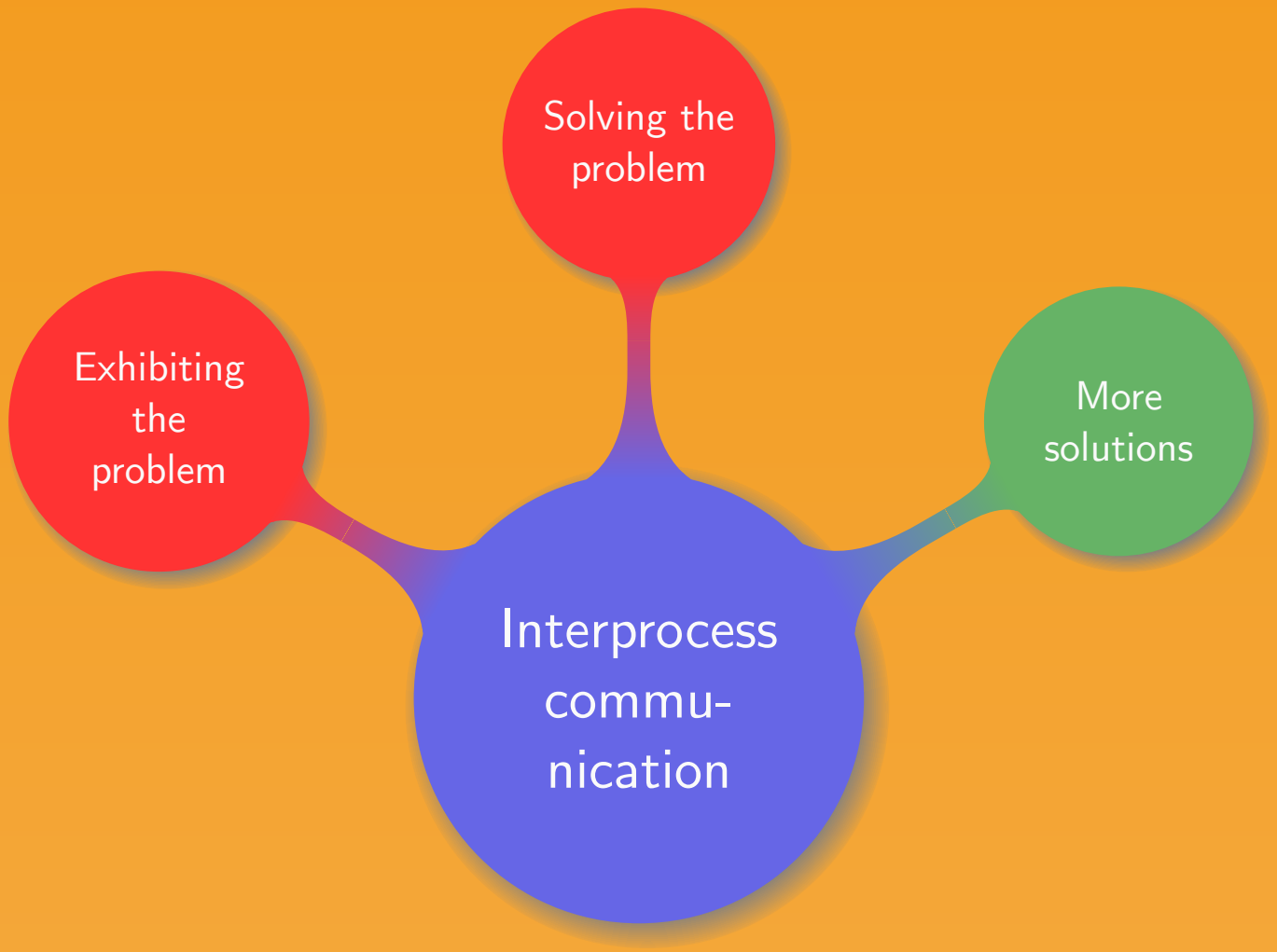
wake the producer

Example

Alter the previous program such as:

- To display information on the consumer and producer
- To increase the buffers to 100
- To have two consumers and one producer. In this case also print which consumer is active.

Chapter organisation



Dangers...

```
1  mutex mut = 0; semaphore empty = 100; semaphore full = 0;
2  void producer() {
3    while(TRUE) {
4      item = produce_item();
5      mutex-lock(&mut);
6      down(&empty);
7      insert_item(item);
8      mutex-unlock(&mut);
9      up(&full);
10   }
11 }
12 void consumer()  {
13   while(TRUE) {
14     down(&full);
15     mutex-lock(&mut);
16     item = remove_item();
17     mutex-unlock(&mut);
18     up(&empty); consume_item(item);
19   }
20 }
```

deadlock.
producer locked & sleeping
consumer cannot access

Never block inside lock

solution: 把6，放在5前面

Questions

In the previous code:

- What is the behavior of the producer when the buffer is full?
- What about the consumer?
- What is the final result for this program?
- How to fix it?

Monitors

As semaphores and mutexes are sometimes risky *monitors* were introduced:

- Monitors are higher level, cleaner and less risky solution
- A monitor is composed of
  - Procedures/functions
  - Structures
- Only one process can be active within a monitor at a time
- Monitors are a programming concept, compiler must know them
- Mutual exclusion handled by the compiler not the programmer

Monitors

Basics on monitors:

- Monitors are easy to use for mutual exclusion
- Offer a condition to block "properly" when the buffer is full
- A signal on the condition variable can wake up a blocked process
- Only one process can be active in the monitor
- As soon as the signal on the condition is sent, exit the monitor
- Other process can resume

Example

```
monitor ProducerConsumer {
  condition full, empty;
  int count;
  void insert(item) {
    if (count == N) wait(full);
    insert_item(item);
    count++;
    if (count==1) signal(empty);
  }
  void remove() {
    if (count==0) wait(empty);
    removed = remove_item;
    count--;
    if (count==N-1) signal(full);
  }
  count:= 0;
}
```

```
void ProducerConsumer::producer() {
  while (TRUE) {
    item = produce_item();
    ProducerConsumer.insert(item);
  }
}
void ProducerConsumer::consumer() {
  while (TRUE) {
    item=ProducerConsumer.remove();
    consume_item(item)
  }
}
```

The compiler knows when to lock

Message passing

many computers conncected together to form a system

Limitation of semaphores and monitors: processes need to share some part of the memory. Distributed systems over a network consist of multiple CPU each with its own private memory

Message passing:

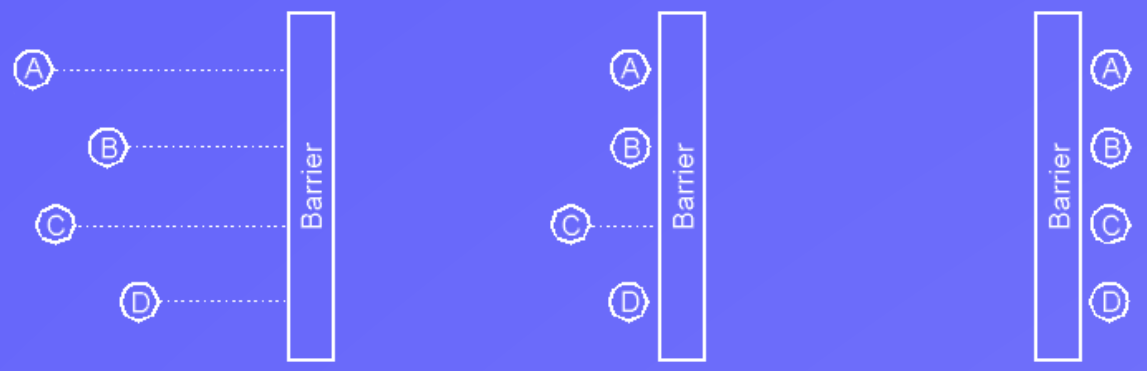
- `send(destination,&message)`
- `receive(source,&message)`, can either block or exit if nothing is received

Potential issues:

- Message lost (sending/acknowledging reception)
- No possible confusion on process names
- Security (authentication, traffic)
- Performance

Barriers

wait for all of the processes to reach the barrier



Useful for problems where several processes must complete before the next phase can start

Key points

- Why is thread communication essential?
- What is a critical region?
- Do software solutions exist?
- What is an atomic operation?
- What are the two best and most common solutions?

Thank you!